

MBDSDR: An AI-Defined Software-Defined Radio Stack with a Function-Calling, Reproducible Measurement Backbone

PENDING_AUTHOR Name, *PENDING IEEE membership* and Bi4MIB open-source contributors

Abstract—We present MBDSDR, a full-stack AI-defined software-defined radio (SDR) reference platform that couples a C++ real-time receive chain, a Flutter mobile client, a Python signal-processing kernel, and an OpenAI-compatible function-calling tool layer exposing every radio operation as an agent-callable tool. This letter focuses on the reproducible experimental backbone: a fixed-seed Monte-Carlo runner with Wilson-score 95% confidence intervals, an E_b/N_0 -in-band-SNR calibration convention, and per-run manifests locking script, seed, data-origin, parameters, sample count, git SHA and UTC timestamp. [SYNC: ten synthetic experiments, $\approx 10.4k$ trials; BPSK sample-rate invariance 0.595–0.705 at fixed E_b/N_0 ; residual ≈ 10 Hz Doppler collapses uncompensated decoding to 6.5% while known-frequency restoration holds ≈ 0.94 ; KNN-AMR 0.96–1.00 vs. classical rules 0.67–1.00 at ≥ 10 dB; local tool pipeline handles 400 mixed calls at 1.000 graceful rate. LLM-agent arm = PENDING_ONLINE_RUN.]

Index Terms—Software-defined radio, AI-defined radio, function calling, automatic modulation recognition, Doppler compensation, reproducible experiments.

I. INTRODUCTION

Software-defined radio (SDR) has moved modulation, filtering and decoding into software, but mainstream SDR tooling [1], [2], [3] still requires manual, expert-driven configuration. Large language models (LLMs) with tool-calling now make it plausible to drive a radio by natural language, yet prior AI-SDR work largely wraps existing GUI software rather than exposing a deterministic, reproducible measurement backbone.

- Layered stack; every SDR op registered as an OpenAI function-calling tool [4].
- Fixed-seed Monte-Carlo library (experiments/common/) with Wilson 95% CIs [5], explicit E_b/N_0 calibration, per-run manifests.
- Ten synthetic experiments, each labelled with origin / N / UTC date.
- Honest separation: deterministic local tool pipeline (measured) vs. online-LLM decision layer (reported as pending).

II. SYSTEM DESIGN

The C++ real-time chain records standard .sigmf-data / .sigmf-meta files under the open SigMF recording

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PENDING_AFFIL: The authors are with [affiliation TBA], [address TBA] (e-mail: PENDING_EMAIL).

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TABLE I
MBDSDR LAYERS (REPRESENTATIVE COMPONENTS).

Layer	Language/location	Role
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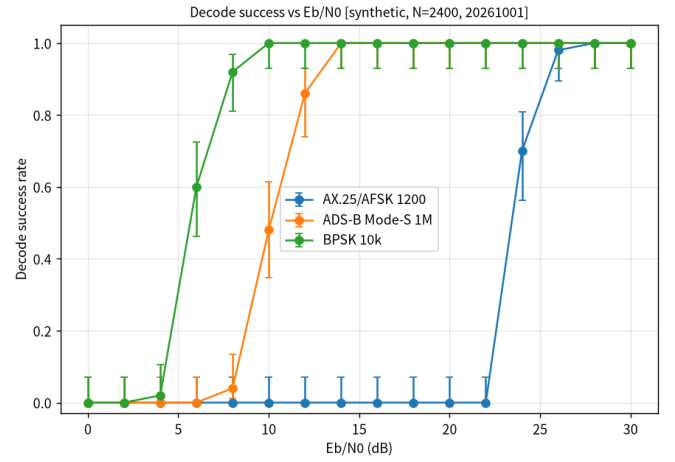


Fig. 1. [Fig.1] Decode success vs. E_b/N_0 (BPSK/ADS-B/AX.25; synthetic).

metadata format [6]. The Python kernel reuses modems and decoders (AX.25/AFSK, ADS-B Mode-S, coherent BPSK, KNN-AMR) unchanged by the experiments, and its orbit-determination code builds on the SGP4/SDP4 element-propagation model [7], [8].

III. EXPERIMENTS

All experiments are synthetic, fixed-seed (master seed 20261001), and run on a headless CI VM with no radio hardware; confidence intervals are Wilson 95% CIs [5].

- Decode success vs. E_b/N_0 (Fig. 1, $N = 2400$)
- Decode success vs. sample rate (Fig. 2, $N = 1400$)
- Decode success vs. receiver bandwidth (Fig. 3, $N = 900$)
- Doppler compensation benefit (Fig. 4, $N = 3600$)
- Automatic modulation recognition (Figs. 5–7, $N = 600/300/640$)
- Single-station Doppler orbit determination (Figs. 8–9, $N = 155/6$)

We propagate a fixed public ISS TLE through SGP4 [7], [8], add fixed-seed Gaussian noise to the Doppler rates observed

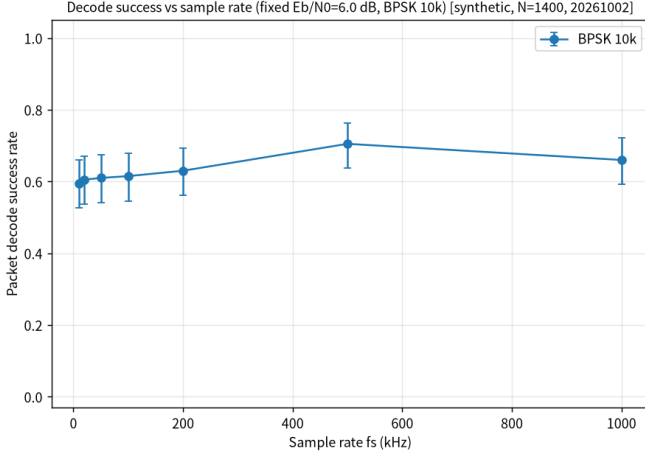


Fig. 2. [Fig.2] Decode success vs. sample rate at fixed $E_b/N_0=6$ dB (synthetic, $N = 1400$).

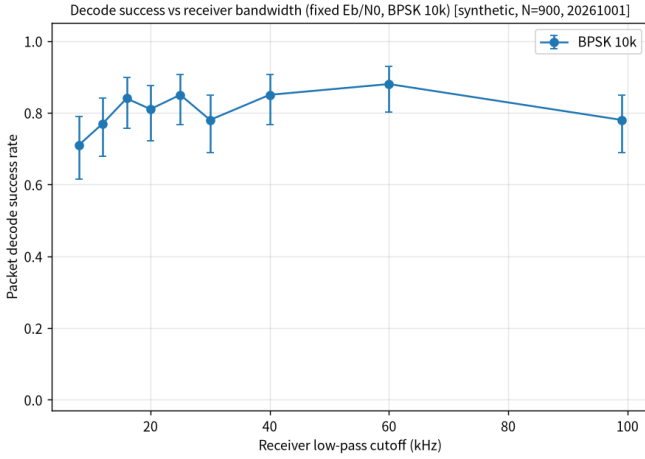


Fig. 3. [Fig.3] Decode success vs. receiver bandwidth (synthetic, $N = 900$).

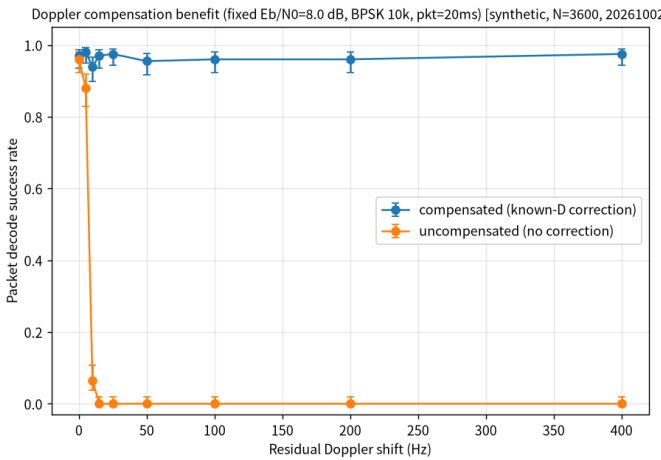


Fig. 4. [Fig.4] Doppler-compensation benefit vs. residual frequency offset (synthetic, $N = 3600$; known-fd compensator = upper bound).

from a single station, and run an EKF and a reference-epoch RLS.

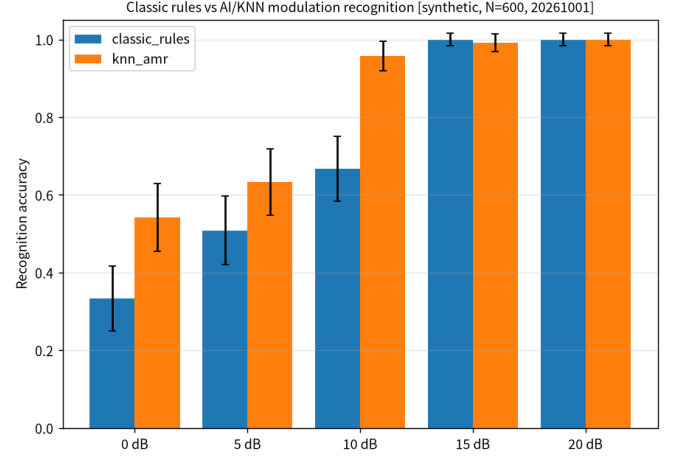


Fig. 5. [Fig.5] Classical-DSP rules vs. KNN-AMR on identical samples (synthetic, $N = 600$; LLM arm not plotted — PENDING).

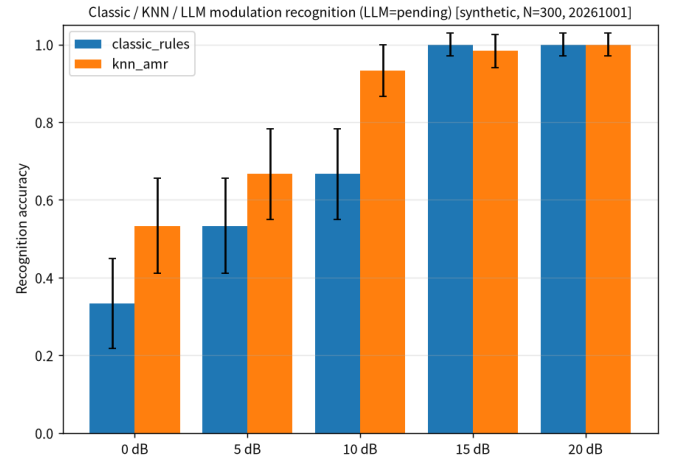


Fig. 6. [Fig.6] AMR SNR grid, same classic/KNN curves; the LLM-agent column is `PENDING_ONLINE_RUN` (`llm_calls_this_run = 0`, no API key in CI) — no LLM accuracy is plotted.

G. Agent tool-call pipeline (Fig. 10, $N = 400$)

IV. DISCUSSION AND LIMITATIONS

V. RELATED WORK

SDR frameworks. Open-source SDR software such as GNU Radio [1], SDR++ [2], and SatDump [3] provides powerful fixed-function receivers but requires manual, expert-driven configuration.

LLM tool-calling and radio agents. Modern chat-completion APIs expose function-calling [4], and open protocols such as the Model Context Protocol [9] connect LLMs to external tools. The closest LLM-driven radio systems we located are preprints on autonomous multi-agent RF signal generation [10] and agentic observability/control of Open-RAN [11]; neither is confirmed peer-reviewed, and both target signal generation or RAN control rather than an open, reproducible SDR measurement backbone.

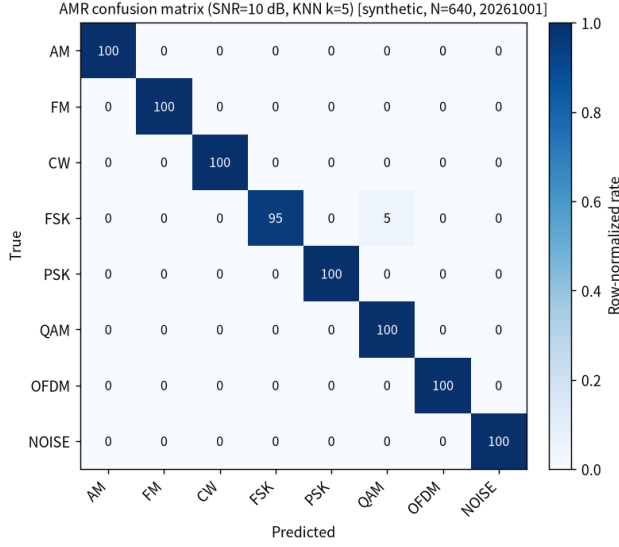


Fig. 7. [Fig.7] Full 8-class AMR confusion matrix at 10 dB (synthetic, $N = 640$; exp_amr.py, independent of the 3-class Fig.5/6 comparison).

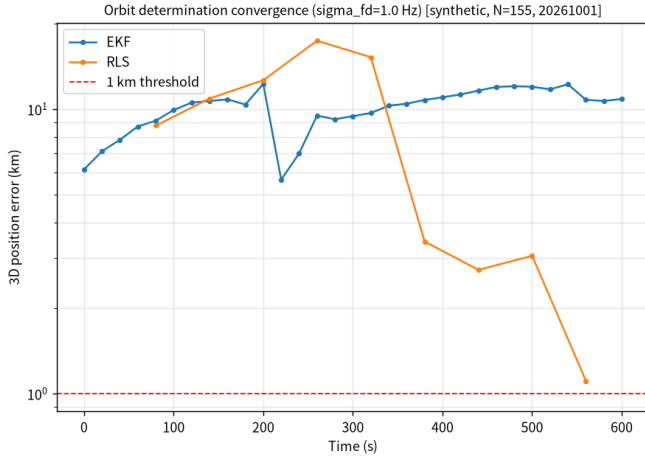


Fig. 8. [Fig.8] Single-station Doppler orbit convergence (synthetic, $N = 155$; $\sigma = 1$ Hz observation noise).

O-RAN / 6G RAN intelligence. Near-RT RIC and xApp surveys [12] apply AI to radio-resource allocation, not to open, reproducible SDR experimentation.

Foundations reused here. Recordings follow the SigMF metadata standard [6]; orbit propagation uses SGP4/SDP4 as originally specified by Hoots and Roehrich [7] and as revisited by Vallado et al. [8]; and all binomial success intervals use the Wilson score interval [5].

This work differs by (a) treating every radio operation as a versioned, schema'd tool, and (b) shipping a fixed-seed, CI-runnable measurement library that reports CIs, data-origin, and per-run manifests so that “AI runs the radio” claims are auditable.

VI. CONCLUSION

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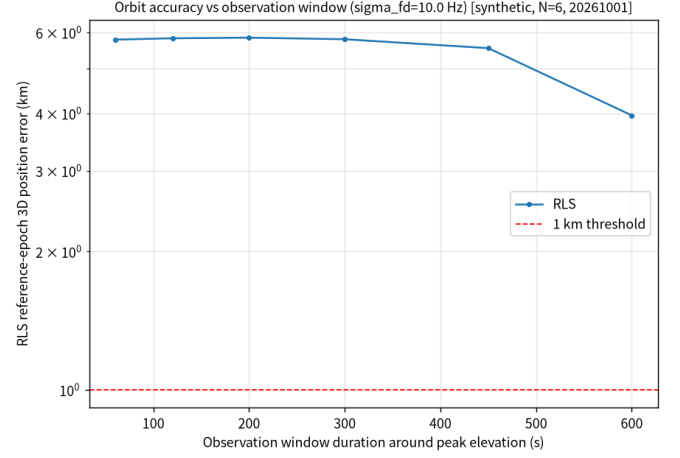


Fig. 9. [Fig.9] Convergence vs. observation-window duration around peak elevation (synthetic, $N = 6$; $\sigma = 10$ Hz).

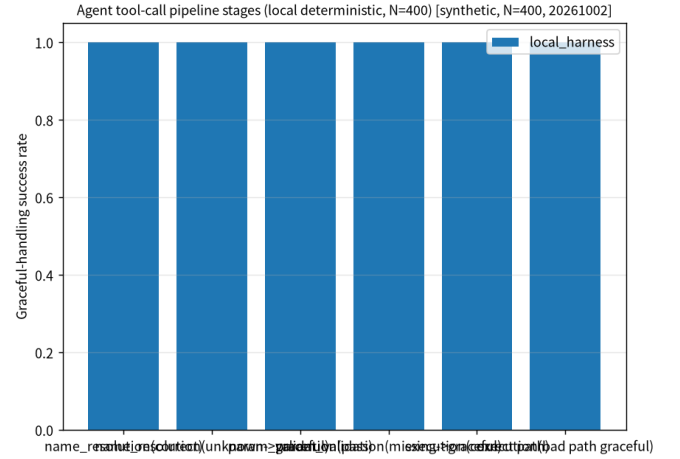


Fig. 10. [Fig.10] Local agent tool-call pipeline stage success (synthetic, $N = 400$; deterministic local stages at 1.000; LLM decision row PENDING).

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